R.V. COLLEGE OF ENGINEERING®

OBSERVATION / DATA SHEET

Date 28.11.23 Name SHREYA J PATIL

Dept./Lab PHYSICS LAB Class _____ Expt./No. 4

Title TRANSISTOR CHARACTERISTICS (EXPEVES-17)

Aim: TO plot the V-I characteristics of the transistor and calculate the parameters α and β

Apparatus: Transistor SL100, 1k and 100k Ω resistors, Bread board, Connecting wires, avegc. kit.

Formulation:

Circuit diagram:



* file not saving on disk!

R.V.C.E. DEPARTMENT EVALUATION SHEET		
Scheme of Evaluation	Course Outcomes	Marks
Data Sheet and Experimentation		
Conclusion		
Viva-Voce		
Total Marks		

4½ 10
10
9
29/30

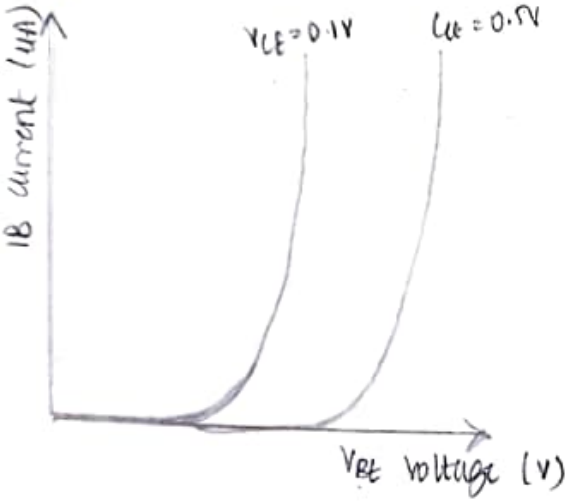
10/28/11

10/05/23

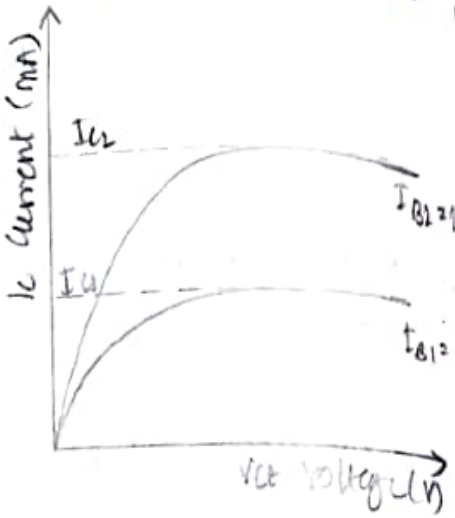
Signature of Teacher Incharge

Model Graphs:

Input characteristics



Output characteristics



Tabular Column:

Input characteristics				Output characteristics		
V_{BE}(V)	I_B(uA)	V_{BE}(V)	I_B(uA)	V_{CE}(V)	I_C(mA)	V_{CE}(V)
0.39V	0.1uA	0.37V	0.1uA	0.1V	0.2uA	0.1V
0.44V	0.21uA	0.45V	0.21uA	0.2V	0.48uA	0.2V
0.48V	0.42uA	0.5V	0.42uA	0.55V	0.5uA	1.03V
0.5V	1.02 1.1uA	0.54V	1:1 1.1uA	1.04V	0.51uA	2.01V
0.52V	1.89uA	0.55V	1.21uA	2.03V	0.52uA	2.85V
0.54V	5.15uA	0.58V	5.57	3.03V	0.52uA	0.54V

Graph plotting with GNUPLOT

Go to respective folder in which the file is saved.
Right click at the empty area and select 'open'
Run the following commands.

→ gnuplot
set timestamp

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OBSERVATION / DATA SHEET

Date 28.11.23 Name SHREYA J PATIL

Dept./Lab PHYSICS LAB Class _____ Expt./No. 4

Title TRANSISTOR CHARACTERISTICS

set title 'Transistor characteristics - Raj'
set xlabel 'Voltage in volts'
set ylabel 'Current in A'
plot 'transistor-input.txt' w lp
plot 'transistor-output.txt' w lp.

Calculations: To calculate transistor parameters α and β from output characteristics

$$\beta = \left[\frac{I_{C_2} - I_{C_1}}{I_{B_2} - I_{B_1}}\right]_{V_{CE}}$$

$$= \frac{(2.552 - 0.552)}{(13.789 - 4.891)} = \frac{1.69}{8.876 \times 10^{-3}} = 190.4$$

Using value of β,

$$\alpha = \frac{\beta}{\beta + 1}$$

$$\alpha = \frac{\beta}{\beta+1} = \frac{190.4}{191.4} = \frac{194.2}{195.2} = 0.9948$$

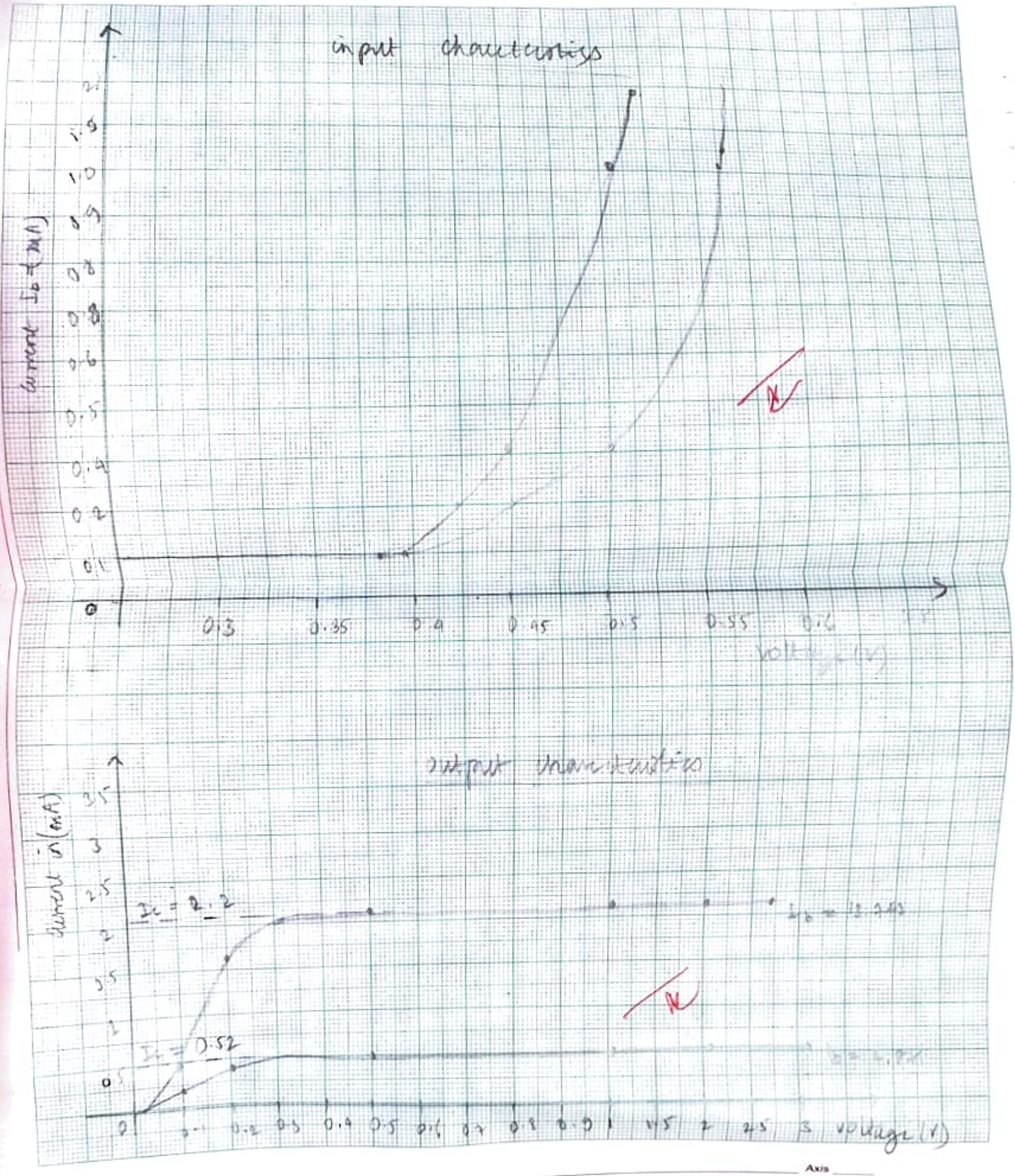
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$$\alpha = \frac{\beta}{\beta+1} = \frac{190.4}{191.4} = 0.9947$$

Result: Current gain (β) in Common emitter configuration is 190.4.

Current gain (α) in Common Base configuration is 0.9948

The input and output characteristics have been plotted and the α and β values are 0.9948 and 194.9.


input characteristics
current I_B (mA)
0.3
0.35
0.4
0.45
0.5
0.55
0.6
output characteristics
current I_C (mA)
I_B = 2.2
I_B = 0.52
voltage (V)

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OBSERVATION / DATA SHEET

Date 28.11.23 Name SHREYA J PATIL

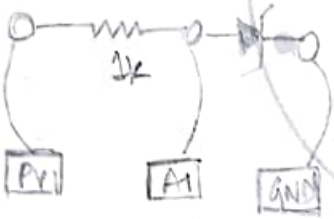
Dept./Lab PHYSICS LAB Class _____ Expt./No. 5

Title ZENER DIODE CHARACTERISTICS.

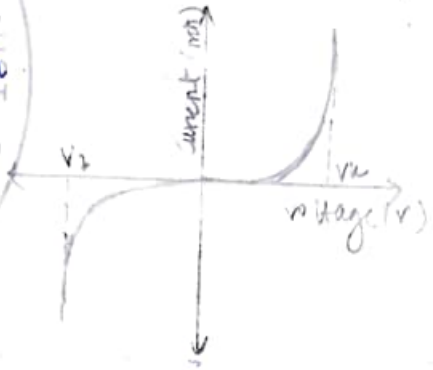
Aim: To study forward and reverse bias characteristics of a zener diode and hence determine forward Bias resistance, knee voltage and zener breakdown voltage.

Apparatus: enpEASE-17 hardware, zener diode, resistor and wires.

Circuit diagram:



Model graph



Scheme of Evaluation	Course Outcome	Max. Marks
Data Sheet and Experimental Setup (Tab)	CO2	4
Conduction of Experiment	CO2	10
Substitution, Calculation & Accuracy	CO3	10
Total Mark's Scored		30
Signature of the Faculty		

4+6 K
10 K
8 K
28/30 K
05/12

K 28/4

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Forward Bias Fit :

x - intercept = 7.35e-01
y - intercept = -4.86e+01
Slope = 6.61e+01

$6.61 \times 10^1 = 66.1$

Reverse Bias Fit :

x - intercept = -2.36e+00
y - intercept = 7.90e+00
Slope = 3.34e+00

Graph plotting with GNUPLOT

Go to respective folder in which the file is saved
Right click at the empty area and select 'open in terminal'.
Run the following commands.

```
set timestamp
set zeroaxis
set title 'Zener Diode characteristics - Rajesh'
set xlabel 'Voltage in volts'
set ylabel 'Current in mA'
plot 'zener.txt' w lp
```

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OBSERVATION / DATA SHEET

Date 29.11.23 Name SHREYA J PATIL

Dept./Lab PHYSICS LAB Class _____ Expt./No. 5

Title ZENER DIODE CHARACTERISTICS

Calculations:

the forward resistance is calculated by,

$$R_f = \frac{\Delta V_f}{\Delta I_f} = \frac{1}{\text{slope}} = \frac{1}{6.61 \times 10^3} = 15.12\ \Omega$$

where ΔV_f is change in forward voltage, ΔI_f is change in forward current in Ampere.

Results:

the knee voltages : 7.35 V

the breakdown voltage : 2.36 V

the forward bias resistance : 15.12 Ω

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OBSERVATION / DATA SHEET

Date 5.12.23 Name SHREYA J PATIL

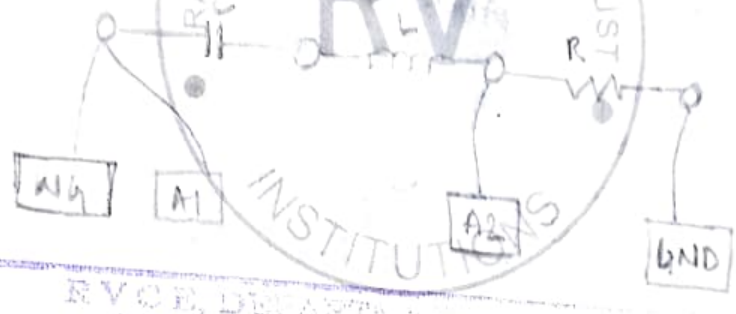
Dept./Lab PHYSICS LAB Class _____ Expt./No. 6

Title SERIES LLR CIRCUIT

Aim: To study the frequency response of LLR circuit and determination of a) self inductance of the given coil. b) Quality factor (Q value) and c) Band-width.

Apparatus: Exp Eyes - A kit, Resistance box, Capacitor, Inductor

Circuit:



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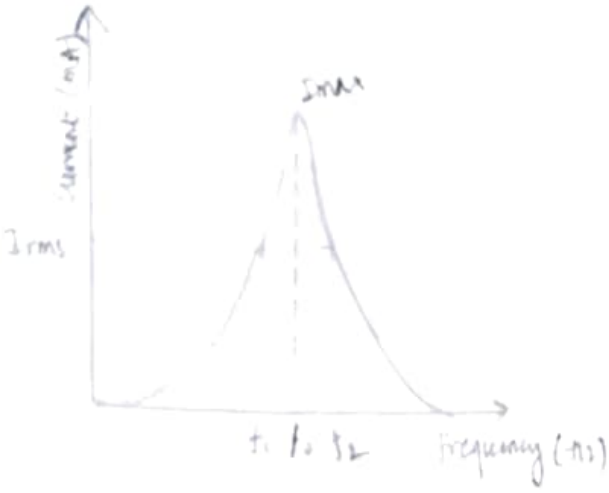
EVCE, DEPARTMENT EVALUATION OF [illegible]		
Scheme of Evaluation	Course Outcome	Max Marks
Data Sheet and Experimental Setup (Neat)	CO1	10
Conduction of Experiment	CO2	10
Substitution, Calculation & Accuracy	CO3	10
Total Marks Scored		30
Signature of the Faculty		

4+6 ✓
9 ✓
9 ✓
28/30 ✓

28/12/23 ✓ 05/12

Signature of Teacher Incharge

Model Graph:



formula:

Imax = 28.206 mA

Irms = Imax/√2 = 19.944 mA

Resonant frequency fo = 211.9 Hz

f1 lower cut off frequency = 186.358 Hz

f2 Upper cut off frequency = 228.033 Hz

Band width Δf = f2 − f1 = 228.033 − 186.358 = 41.675

Quality factor Q = fo/Δf = 5.08

Capacitance C = 474 nF

$L = \frac{1}{4\pi^2 f_o^2 C}$ = H

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OBSERVATION / DATA SHEET

Date 5.12.23 Name SHREYA J PATIL

Dept./Lab PHYSICS Class _____ Expt./No. 16

Title SERIES LCR CIRCUIT

Graph plotting with GNUPLOT

Go to respective folder in which the file is saved

Right click at the empty area, select 'Open in terminal' and Run the following commands.

Set timestamp

Set title 'Series LCR circuit'

Set xlabel 'Frequency in Hz'

Set ylabel 'Current in mA'

plot 'LCR.txt' w lp

making points on the graph.

- Take mouse to the tip of the curve
- Press the scroll button on the mouse
- Note down the resonant frequency and Imax
- Find out Ioms
- Press the scroll button on the mouse at Imax
- Note down f_1 and f_2
- Export the file as PDF.

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Calculation:

$$L = \frac{1}{4\pi^2 f_0^2 C}$$

$$= \frac{1}{4 \times (3.14)^2 \times (211.988)^2 \times 474 \times 10^{-9}}$$

$$= \frac{1}{4 \times 9.8596 \times 44921.9547 \times 474 \times 10^{-9}}$$

$$= \frac{1}{0.8397}$$

$$= 1.190 \text{ H}$$

RESULT:

1. Resonant frequency of the circuit f_0	211.988	Hz
2. Self-inductance of the given coil L	1.190	H
3. Quality factor from graph Q	5.08	
4. Band width $\Delta f = f_2 - f_1$	41.675	Hz

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OBSERVATION / DATA SHEET

Date 5.12.23 Name SHREYA J PATIL

Dept./Lab PHYSICS LAB Class _____ Expt./No. 7

Experiment: Wavelength of light emitting diodes (expEYES-17)

Aim: To study the I-V characteristics of a diode and determine the wavelength of the given LED's

Apparatus: expEYES-17 hardware, LED's, 1kΩ Resistor and connecting wires.

Formula: Energy of the photons emitted by LED $\Rightarrow E = \frac{hc}{\lambda} = eV_k$

the Wavelength of LED is $\lambda = \frac{hc}{eV_k}$ nm.

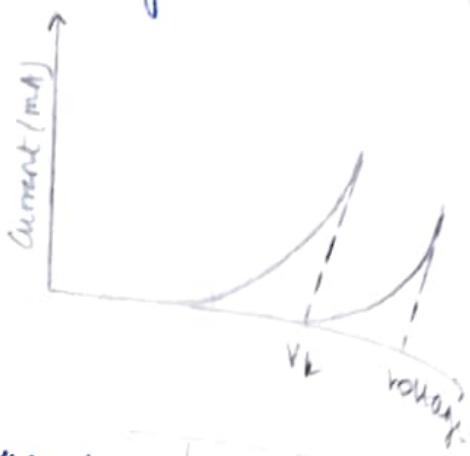
RV COLLEGE OF ENGINEERING EVALUATION CHART			
Scheme of Evaluation	Course Outcome	Max Marks	
Data Sheet and Experimental Setup (4+6)	CO1	10	4+6
Conduction of Experiment	CO2	10	9
Calculations, Graph & Accuracy	CO3	10	10
Total Marks Scored		30	29/30
Signature of the Faculty			

Signature of Teacher Incharge

Circuit diagram



model Graph



Color of LED	knee voltage (V_k)	Wavelength (λ)
Red	1.72	722
Blue	2.54	489
Yellow	1.83	592
Green	2.23	557

Graph plotting with GNUPLOT

Go to respective folder in which the file is saved

Right click at the empty area and select 'open in terminal'.

Run the following commands

```
gnuplot
set timestamp
set title 'wavelength of LED'
set xlabel 'voltage in volts'
set ylabel 'current in mA'
plot 'led.txt' w lp.
```

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OBSERVATION / DATA SHEET

Name SHREYA J PATIL

5.12.23

Lab PHYSICS LAB Class _____ Expt./No. 7

Wavelength of LED

For red light : $\lambda = \frac{hc}{eV_R}$

Calculations :

$$= \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19} \times 1.72} = \frac{1.9878 \times 10^{-25}}{2.752 \times 10^{-19}} = 722 \text{ nm}$$

Blue light : $\lambda = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19} \times 2.54} = \frac{1.9878 \times 10^{-25}}{4.064 \times 10^{-19}} = 489 \text{ nm}$

Yellow light : $\lambda = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19} \times 1.83} = \frac{1.9878 \times 10^{-25}}{2.928 \times 10^{-19}} = 591 \text{ nm}$

Green light : $\lambda = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19} \times 2.23} = \frac{1.9878 \times 10^{-25}}{3.568 \times 10^{-19}} = 557 \text{ nm}$

Result : Wavelength of given LED's are determined.

05/12/23

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OBSERVATION / DATA SHEET

ate 21.11.2023 Name SHREYA J PATIL

ept./Lab PHYSICS LAB Class _____ Expt./No. 3

tle LASER DIFFRACTION

m: To determine the wavelength of a given laser Beam.

paratus: laser source, Grating, Optical bench with accessories and metre scale etc.

rmula: Wavelength of laser light, $\lambda = \frac{C \sin\theta_n}{n} \quad \ldots m$

where C is the grating constant and is the distance between corresponding points of two successive lines on the grating, n is the order of the maximum, θ_n - angle of diffraction of the nth maximum.

Grating Constant $C = \frac{1 \text{ inch}}{\text{no of lines (N) per inch}} = \frac{2.54 \times 10^{-2} m}{500} = 5.08 \times 10^{-5} m$

Distance between grating and the screen, d = 1410 cm.

HPG
10
9
29

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Diagram:

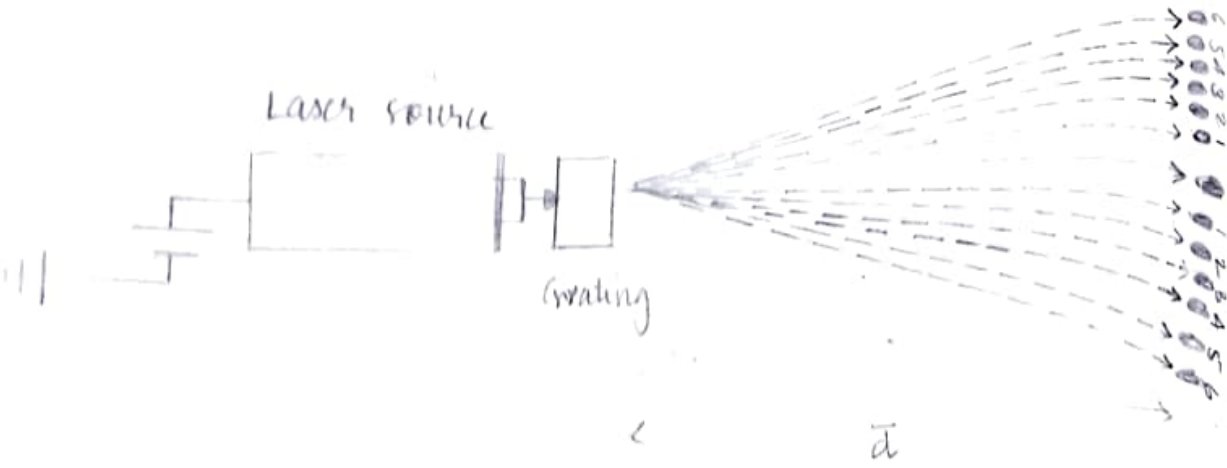


TABLE Table:

Diffraction order (n)	distance $2x_n$ (cm)	distance x_n (cm)	diffraction angle (θ_n) $\theta = \tan^{-1}\left(\frac{x_n}{d}\right)$	wavelength $\lambda = \frac{(s$
1	3.0	1.5	0.7734	680
2	5.8	2.9	1.4955	660
3	8.8	4.4	2.2677	668
4	11.8	5.9	3.0395	673
5	14.5	7.25	3.7361	661
6	17.4	8.7	4.4771	660

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OBSERVATION / DATA SHEET

Date 01 11 23 Name SHREYA J PATIL

Dept./Lab PHYSICS LAB Class _____ Expt./No. 3

Title LASER DIFFRACTION

Calculations:

$n = 1$: $\sin\theta_1 = 0.0134$

$$\lambda_1 = \frac{5.08 \times 10^{-5} \times 0.0134}{1} = 6.804 \times 10^{-7}$$

$n = 2$: $\sin\theta_2 = 0.0260$

$$\lambda_2 = \frac{5.08 \times 10^{-5} \times 0.0260}{2} = 6.604 \times 10^{-7}$$

$n = 3$: $\sin\theta_3 = 0.0395$

$$\lambda_3 = \frac{5.08 \times 10^{-5} \times 0.0395}{3} = \frac{2.0066 \times 10^{-6}}{3} = 6.68 \times 10^{-7}$$

$n = 4$: $\sin\theta_4 = 0.0530$

$$\lambda_4 = \frac{5.08 \times 10^{-5} \times 0.0530}{4} = \frac{2.6924 \times 10^{-5}}{4} = 6.73 \times 10^{-7}$$

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$n = 5 \qquad \sin\theta_5 = 0.0651$

$$\lambda_5 = \frac{5.08 \times 10^{-5} \times 0.0651}{5} = \frac{3.3070 \times 10^{-6}}{5} = 6.61 \times 10^{-7}$$

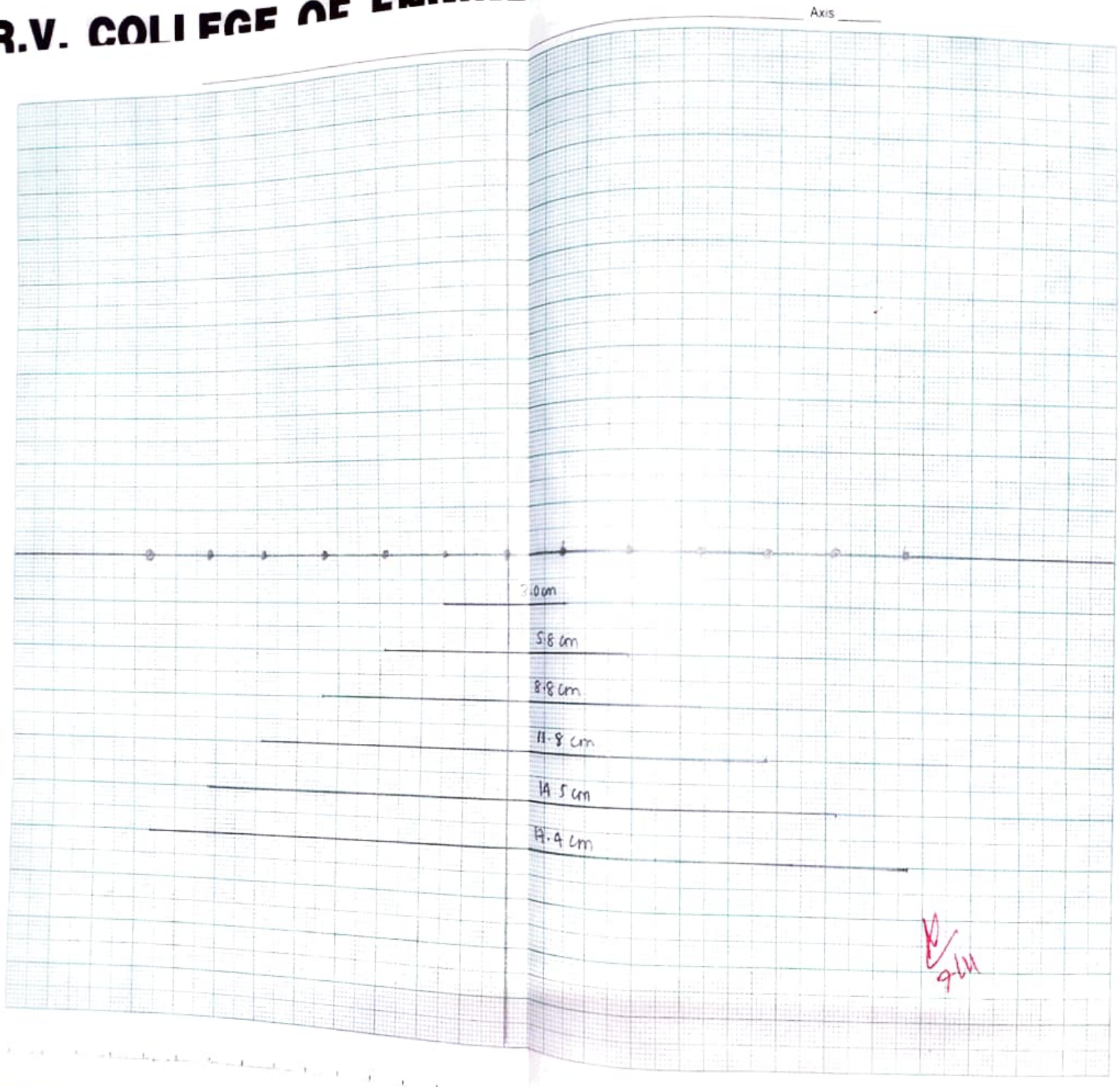
$n = 6 \qquad \sin\theta_6 = 0.0780$

$$\lambda_6 = \frac{5.08 \times 10^{-5} \times 0.0780}{6} = \frac{3.9624 \times 10^{-6}}{6} = 6.60 \times 10^{-7}$$

Average value of $\lambda = \frac{6.80 + 6.60 + 6.65 + 6.73 + 6.61 + 6.60}{6}$

$= 6.49 \times 10^{-7}$ cm

Result : The wavelength of the laser light is found to be 6.49×10^{-7} cm $= 649$ nm


Axis
Axis
5.8 cm
8.8 cm
11.8 cm
14.5 cm
17.4 cm

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OBSERVATION / DATA SHEET

Date 19.12.23 Name SHREYA J PATIL

Dept./Lab PHYSICS LAB Class _____ Expt./No. 9

Title BLACK BOX

Aim: Identification of the unknown passive electrical components (L, C and R) enclosed in a black box and determination of their values.

Apparatus: Black box, audio frequency oscillator, ac milli-ammeter (0-20mA) and ac voltmeter (digital multimeter in ac voltage mode).

Formulae:

- For resistance: $R = \frac{V}{I} (\Omega)$
- For inductance: $L = \frac{V}{2\pi f I} (H)$

f - frequency of applied signal
V - potential difference
I - current
L - inductance
C - capacitance
R - Resistance.

Scheme of Evaluation	Course Outcome	Max. Marks	
Observation Sheet and Experimental Setup (10)	CO1	10	9
Conduction of Experiment	CO2	10	9
Tabulation, Calculations and Accuracy	CO3	10	9
Result and Discussion		5	4
Viva-voce and Punctuality		5	4
		40	36/40

02/01/24

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TERMINALS C_1 and C_2

Frequency (Hz)	voltage (V)	Current I (mA)	Component value (H) unit ?
100	6.9 V	32.6 A	0.337
200	7.3	21.1	0.275
300	7.6	14.2	0.284
400	7.6	10.6	0.285
500	7.6	7.9	0.306
600	7.6	6.2	0.325

Terminals C_1 and C_2 correspond to Inductor.

the avg value of the component is 0.302

CALCULATIONS:

For Resistance: $R = \frac{V}{I} (\Omega)$

$R = \frac{9.5}{1.5} = 6.33$ $\quad R = \frac{8.9}{0.7} = 12.71$

$R = \frac{9.2}{0.7} = 13.14$

$R = \frac{9.1}{0.7} = 13$

$R = \frac{9.0}{0.7} = 12.85$

$R = \frac{8.9}{0.7} = 12.71$

$\text{avg } R = \frac{6.33 + 13.14 + 13 + 12.85 + 12.71 +}{6}$

$= 70.74$

$= 11.79$

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OBSERVATION / DATA SHEET

Date 19.12.23 Name SHREYA J PATIL

Dept./Lab PHYSICS LAB Class _____ Expt./No. 9

Title BLACK BOX.

For inductance : $L = \frac{V}{2\pi f I}$ (H)

$\frac{6.9}{2 \times 3.14 \times 100 \times 32.6 \times 10^{-3}} = 0.337$ H $\quad L = \frac{7.6}{2 \times 3.14 \times 400 \times 10.6 \times 10^{-3}} = 0.285$

$\frac{7.3}{2 \times 3.14 \times 200 \times 21.1 \times 10^{-3}} = 0.275 \quad L = \frac{7.6}{2 \times 3.14 \times 500 \times 7.9 \times 10^{-3}} = 0.306$

$\frac{7.6}{2 \times 3.14 \times 300 \times 14.2 \times 10^{-3}} = 0.284 \quad L = \frac{7.6}{2 \times 3.14 \times 600 \times 6.2 \times 10^{-3}} = 0.325$

Avg $L = \frac{0.337 + 0.275 + 0.284 + 0.285 + 0.306 + 0.325}{6} = \frac{1.812}{6}$

$= 0.302$

For capacitance : $C = \frac{I}{2\pi f V}$

$\frac{2.3 \times 10^{-3}}{2 \times 3.14 \times 100 \times 9.6} = \frac{0.3815 \times 10^{-6}}{}$ = 0.3815 μF

$\frac{3.9 \times 10^{-3}}{2 \times 3.14 \times 100 \times 9.4} = 0.3303 \times 10^{-6} = 0.3303$ μF

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$$C = \frac{I}{2\pi f V} = \frac{6.1 \times 10^{-3}}{2 \times 3.14 \times 800 \times 9.3} = 0.3481 \times 10^{-6} = 0.3481\ \mu F$$

$$C = \frac{I}{2\pi f V} = \frac{8.2 \times 10^{-3}}{2 \times 3.14 \times 400 \times 9.1} = 0.3571 \times 10^{-6} = 0.3571\ \mu F$$

$$C = \frac{I}{2\pi f V} = \frac{10.8 \times 10^{-3}}{2 \times 3.14 \times 500 \times 9.0} = 0.3644 \times 10^{-6} = 0.3644\ \mu F$$

$$C = \frac{I}{2\pi f V} = \frac{12.4 \times 10^{-3}}{2 \times 3.14 \times 800 \times 9.0} = 0.3656 \times 10^{-6} = 0.3656\ \mu F$$

$$\text{Avg } C = \frac{0.3815 + 0.3303 + 0.3481 + 0.3571 + 0.3644 + 0.3656}{6}$$

$$= \frac{2.147}{6} = 0.3578\ \mu F$$

R_{total}

1. the component Capacitor of value 0.3578 is connected across terminals A_1 and A_2.
2. the component Resistor of value 11.79 is connected across terminals B_1 and B_2.
3. the component Inductor of value 0.302 is connected across terminals C_1 and C_2.

Paste the data sheets here